EEP 595: Privacy-Preserving Machine Learning – Spring 2026

Course Project #1– Project Proposal

Assigned: Wednesday, April 8, 2026, **Due: Saturday, April 25, 2026**

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1 Overview

Over the course of the next nine weeks, you will explore a topic of your choice at **the intersection of privacy and machine learning**. Your project **needs to** combine two equally important components:

- **A literature review component** You are expected to identify, read, and critically engage with relevant research papers. This is not a superficial survey - you should understand the key ideas, situate them in the broader PPML landscape, and be able to discuss their strengths, limitations, and open questions.
- **A practical component** You are expected to complement your literature work with something hands-on - **an implementation, a set of experiments, a simulation, or a prototype system.** Purely theoretical or survey-only projects will not meet the requirements of this course.

Both components should be clearly present in every phase of the project, from the proposal through the final report. *A strong project is one where the practical work is informed by the literature, and the literature is illuminated by the practical work.*

Your projects can be individual or in teams of up to three people. Team size will be factored into expectations - larger teams are expected to produce broader, more comprehensive work.

Possible directions include, but are not limited to:

- Implementing and evaluating a PPML technique (e.g., federated learning, differential privacy, secure multi-party computation, homomorphic encryption) on a real-world dataset.
- Comparing the privacy-utility tradeoff of multiple approaches on a shared problem.
- Identifying a privacy vulnerability in an existing ML pipeline, and proposing/designing a mitigation.
- Reproducing and critically extending the results of a published PPML paper.
- Designing and prototyping a system that applies PPML techniques to a specific application domain (e.g., healthcare, finance, NLP).

2 Part 1 - Team Formation, Project Proposal and Planning

In the next two wees, please form your team, or decide to work alone, and settle on a topic. This project part has two components, which your will want to submit together as a single document.

2.1 Component 1: Project Proposal (2 pages)

Think of your proposal as a pitch to a results-oriented stakeholder - a potential client or executive - who may or may not be deeply familiar with the problem. Get to the point quickly, and make the case that the problem matters and that your approach is sound.

Your proposal should address the following:

- **1. Problem description** Please provide a clear description of the PPML problem you are addressing. If your project focuses on a specific attack or class of attacks, you will likely want to structure your analysis around:
  - **System:** What is it, and how does it work?
  - **Assets:** What are you trying to protect, and why? How valuable are those assets?
  - **Attackers:** Who might attack the system, and what would motivate them?
  - **Vulnerabilities:** Where might the system be weak?
  - **Threats:** What actions might an attacker take to exploit those weaknesses?
  - **Risks:** How likely is exploitation, and how severe would the consequences be?

Diagrams and figures are welcome but not required.

- —bf 2. Proposed approach If your project focuses on preventing or mitigating the described problem, describe your planned approach. Consider addressing:
  - What broad category does your solution fall into (e.g., cryptography, system redesign, differential privacy)?
  - What are the expected benefits of the solution?
  - What are the known or anticipated drawbacks?
- **3. References.** For each paper you are referencing or plan to reference, provide the full title and author list.

2.2 Component 2: Project Plan (1 page)

Alongside your proposal, please submit a one-page project plan. This is not a SCRUM document — it is simply a structured, honest account of what you intend to do and when. It should include:

- **Final product description.** A brief list of the expected deliverables at the end of the project (report, presentation, code, experiments, etc.) and what each will contain.
- **Milestone plan.** A week-by-week breakdown of what you plan to accomplish during this project. Be concrete - vague entries like "work on implementation" are less useful than "complete baseline federated learning prototype and run initial accuracy benchmarks."
- **Responsibilities.** If working in a team, indicate which team member is leading each major workstream.

We will review proposals and return comments promptly so you can begin your work with confidence.

2.3 Grading Criteria

We will grade your proposals based upon:

- Clarity and significance of the research question.
- Quality of threat/problem analysis.
- Demonstrated awareness of relevant prior work.
- Feasibility and concreteness of the plan.
- Appropriateness of scope given team size.

3 A Note on Scope and Ambition

Graduate-level work is expected. That does not mean your project needs to be groundbreaking, but it does mean engaging seriously with the technical and ethical dimensions of your topic. Privacy-preserving ML involves real tradeoffs — between utility and privacy, between practicality and theoretical guarantees, between individual and collective interests. The strongest projects will grapple with these tradeoffs honestly rather than treating them as footnotes.